MILITARY SPECIFICATION

MICROCIRCUITS, DIGITAL, BIPOLAR, LOW POWER SCHOTTKY TTL, DATA SELECTOR/MULTIPLEXER, WITH THREE-STATE OUTPUTS, CASCADABLE, MONOLITHIC SILICON

INACTIVE FOR NEW DESIGN AFTER 18 April 1997

This amendment forms a part of MIL-M-38510/309D, dated 21 June 1985, and is approved for used by all Departments and Agencies of the Department of Defense.

PAGE 1

Title: Delete and substitute as printed above.

1.2.3, add the following case outline:

"X C-2A (20 terminal, .350" x .350", square chip carrier package)"

Page 2

1.3, Thermal resistance, junction-to case (θ_{JC}) and Junction temperature (T_J), delete and substitute:

"Thermal resistance, junction-to-case (θ_{JC})
Cases E, F, X, and 2 ----------See MIL-M-38510, appendix C
Junction temperature (T_J) 3/ ----------+175°C

Footnote 3/, delete in its entirety.

PAGE 3

Following 3.2.3, add:

"3.2.4 Case outline. Case outline shall be in accordance with 1.2.3."

4.2b: Delete "Initial" and substitute "Interim" (2 places) (as substituted in Amendment 1).

PAGE 9

TABLE II, Group A test requirements, class B: Add "8".

TABLE II, Additional electrical subgroups for group C periodic inspections: Delete in its entirety.

4.4.3b.3: Delete "appendix B of MIL-M-38510 and".

PAGES 13, 14, 15, AND 16

FIGURE 1: Delete "Case 2" and substitute "Cases X and 2".

The attached insertable replacement pages listed below are replacements for stipulated pages. When the new pages have been entered in the document, insert the amendment as the cover sheet to the specification.

Replacement page	Page replaced
25	Reprinted without change
26	26

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PAGES 32, 33, AND 34

TABLE III, device type 01, case 2: Delete and substitute "Cases X and 2".

PAGE 35

TABLE III, device type 01, case 2: Delete and substitute "Cases X and 2:.

TABLE III, footnote 1/, I_{IL1}, circuit E: Delete "-.16/-.40" and substitute "-.002/-.150".

TABLE III, footnote 1/, I_{IL2}, circuit E: Delete "-.16/-.40" and substitute "-.002/-.150".

PAGES 36 AND 37

TABLE III, device type 02, case 2: Delete and substitute "Cases X and 2".

PAGE 38

TABLE III, device type 02, case 2: Delete and substitute "Cases X and 2".

TABLE III, footnote 1/, I_{IL}, circuit F: Add "-.120/-.360".

PAGES 39 AND 40

TABLE III, device type 03, case 2: Delete and substitute "Cases X and 2".

PAGE 41

TABLE III, device type 03, case 2: Delete and substitute "Cases X and 2".

TABLE III, footnote 1/, I_{IL1}, circuit E: Delete "-.16/-.40" and substitute "0/-.20".

TABLE III, footnote 1/, I_{IL2}, circuit E: Delete "-.23/-.71" for test 27 and "-.32/-.80" for test 28 and substitute "0/-.10" (2 places).

PAGES 42 AND 43

TABLE III, device type 04, case 2: Delete and substitute "Cases X and 2".

PAGE 44

TABLE III, device type 04, case 2: Delete and substitute "Cases X and 2".

TABLE III, t_{PLH5} and t_{PHL5}: Reenter in its entirety (alters change made in Amendment 3).

TABLE III, t_{PLH5}, Test numbers 88 through 91: All A pin outs, delete "GND" and substitute "5.0 V" (4 places); all B pin outs, delete "5.0 V" and substitute "GND" (4 places).

TABLE III, t_{PHL5}, Test numbers 92 through 95: All A pin outs, delete "5.0 V" and substitute "GND" (4 places); all B pin outs, delete "GND" and substitute "5.0 V" (4 places).

TABLE III, footnote 1/, I_{IL1}, circuit E: Delete "-.16/-.40" and substitute "0/-.20".

TABLE III, footnote 1/, I_{IL2}, circuit E: Delete "-.23/-.71" for test 27 and "-.32/-.80" for test 28 and substitute "0/-.10" (2 places).

PAGE 45

TABLE III, device type 05, case 2: Delete and substitute "Cases X and 2".

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PAGE 46

TABLE III, device type 05, case 2: Delete and substitute "Cases X and 2".

TABLE III, I_{OS}, test number 57, pins D6, D5, D4: Delete "ditto mark" (alters change made in a previous amendment).

PAGE 47

TABLE III, device type 05, case 2: Delete and substitute "Cases X and 2".

PAGE 48

TABLE III, device type 05, case 2: Delete and substitute "Cases X and 2".

TABLE III, t_{PLZ1}, test number 128, maximum limits column: Delete "30" and substitute "35".

TABLE III, t_{PLZ2}, test number 129, maximum limits column: Delete "30" and substitute "35".

TABLE III, footnote 1/, I_{IL2}, circuit E: Delete "-.16/-.40" and substitute "-.002/-.150".

* TABLE III, 1/ I_{IL} Limits table: Delete footnote and substitute as printed below.

1/ I_{IL} limits shall be as follows:

Test	Min/Max limits (mA) for circuit						
	A	B	C	D	E	F	G
I_{IL1}	-.16/-.40	-.12/-.36	-.16/-.40	-.03/-.30	-.005/-.72	-.105/-.345	0/-.15
I_{IL2}	0/-.20	-.12/-.36	-.12/-.36	-.03/-.30	-.002/-15	-.16/-.40	0/-.15
I_{IL3}	-.12/-.36	-.12/-.36	-.12/-.36	-.03/-.30	-.10/-,34	-.16/-.40	0/-.15

PAGES 49 AND 50

TABLE III, device type 06, case 2: Delete and substitute "Cases X and 2".

PAGE 51

TABLE III, device type 06, case 2: Delete and substitute "Cases X and 2".

TABLE III, footnote 1/, delete in its entirety and substitute (alters changes made by Amendment 3):

1/ I_{IL} limits shall be as follows:

Test	Min/Max limits (mA) for circuit						
	A	B	C	D	E	F	G
I_{IL1}	-.15/-.38	-.16/-.40	-.20/-.44	0/-.30	0/-.20	-.12/-.36	0/-.15
I_{IL2} test 27	0/-.20	-.16/-.40	-.20/-.44	0/-.30	0/-.10	-.12/-.36	0/-.15
test 28	0/-.20	-.32/-.80	-.40/-.88	0/-.60	0/-.10	-.24/-.72	0/-.15

PAGES 52 AND 53

TABLE III, device type 07, case 2: Delete and substitute "Cases X and 2".

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PAGE 54

TABLE III, device type 07, case 2: Delete and substitute "Cases X and 2".

TABLE III, subgroup 10, delete and substitute new subgroup 10:

"

10 T_C = +125°C	Same test and terminal conditions as subgroup 9, except T_C = +125°C and limits as follows: t_{PLH1} and t_{PHL1} = 3 to 35 ns; t_{PLH5} and t_{PHL5} = 3 to 39 ns; t_{PZH3}, t_{PZL3}, and t_{PHZ3} = 3 to 53 ns; t_{PLZ3} = 3 to 45 ns.

"

TABLE III, footnote 1/, delete in its entirety and substitute (alters changes made by Amendment 3):

1/ I_{IL} limits shall be as follows:

Test	Min/Max limits (mA) for circuit						
	A	B	C	D	E	F	G
I_{IL1}	-.15/-.38	-.16/-.40	-.20/-.44	0/-.30	0/-.20	-.12/-.36	0/-.15
I_{IL2} test 27	0/-.20	-.16/-.40	-.20/-.44	0/-.30	0/-.10	-.12/-.36	0/-.15
test 28	0/-.20	-.32/-.80	-.32/-.80	0/-.60	0/-.10	-.24/-.72	0/-.15

PAGES 55, 56, AND 57

TABLE III, device type 08, case 2: Delete and substitute "Cases X and 2".

PAGE 58

TABLE III, device type 09, case 2: Delete and substitute "Cases X and 2".

PAGE 59

TABLE III, device type 09, case 2: Delete and substitute "Cases X and 2".

TABLE III, t_{PLH1} and t_{PHL1}, test numbers 73 through 80, V_{CC} column: Delete "5.5 V" and substitute "5.0 V".

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The margins of this amendment are marked with asterisks to indicate where changes from the previous amendment were made. This was done as a convenience only and the Government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and contractors are cautioned to evaluate the requirements of this document based on the entire content irrespective of the marginal notations and relationship to the last previous amendment.

CONCLUDING MATERIAL

Custodians:
Army - CR
Navy - EC
Air Force - 11
DLA - CC

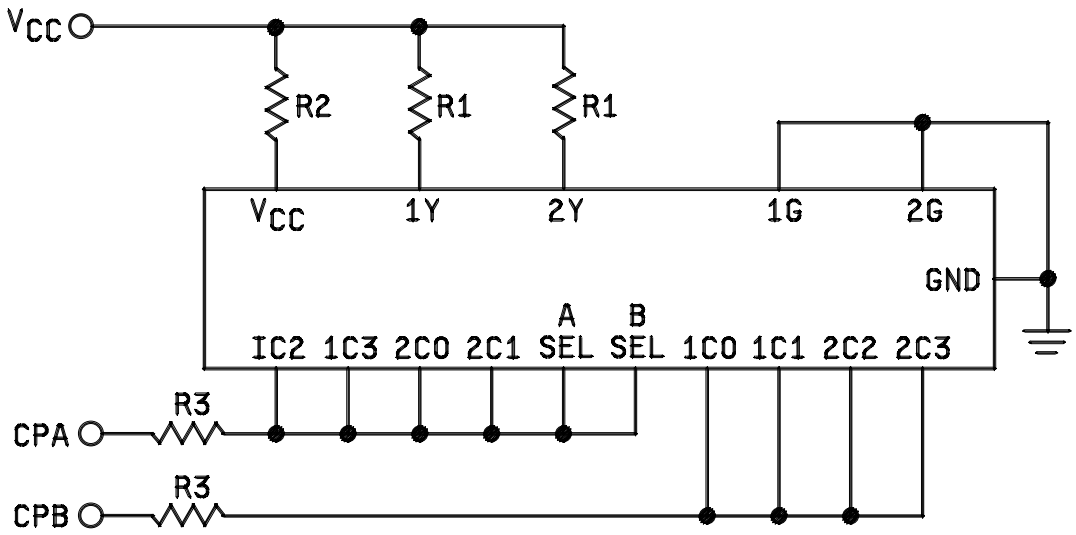
Review activities:
Army - HD, MI, SM
Navy - AS, CG, MC, SH, TD
Air Force - 03, 19, 99

Preparing activity:
DLA - CC

(Project 5962-1923)

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Device types 02 and 08



NOTES:

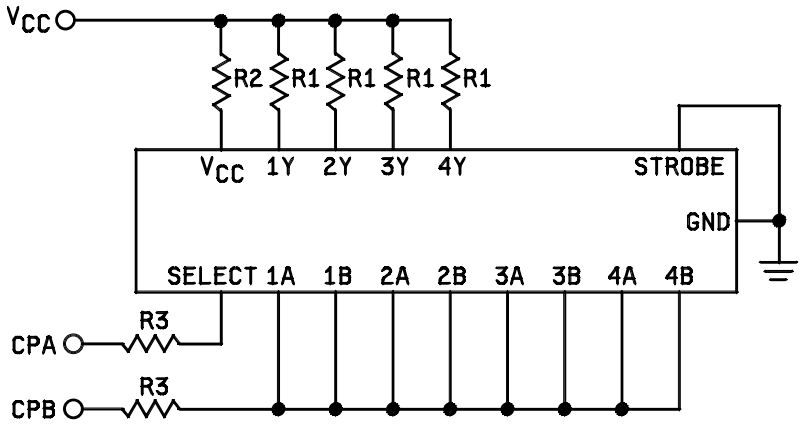
1. CPA = 100 kHz ±50% square wave; duty cycle = 50 ±15%; V_{IL} = -0.5 V minimum to +0.7 V maximum; V_{IH} = 2.0 V minimum to 5.5. V maximum.
   CPB = same as CPA, synchronized with CPA, except 50 kHz ±50% square wave.
2. R1 = 1 kΩ ±5%; R2 = 10Ω ±5%; R3 = 27Ω ±5%.
3. V_{CC} shall be high enough to insure that 5.0 V minimum is present at V_{CC} device terminal.

FIGURE 4. Burn-in and life test circuit - Continued.

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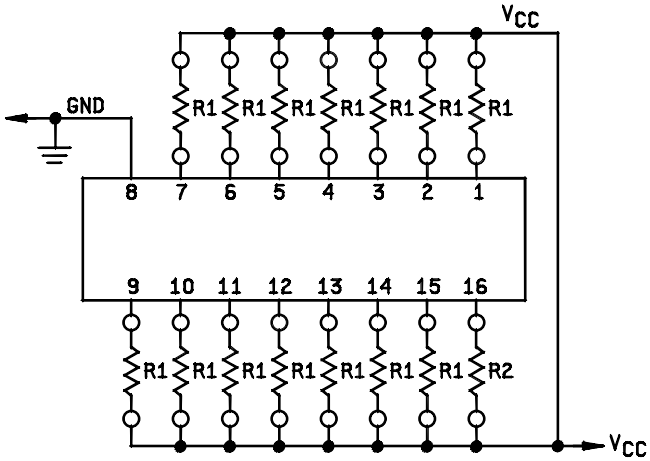
Device types 03, 04, 06, and 07



NOTES:

1. CPA = 100 kHz ±50 percent square wave; duty cycle = 50 ±15 percent; V_{IL} = -0.5 V minimum to +0.7 V maximum; V_{IH} = 2.0 V minimum to 5.5. V maximum. CPB = same as CPA, synchronized with CPA, except 50 kHz ±50 percent square wave.
2. R1 = 1 kΩ ±5 percent for device types 03 and 04
   R1 = 10Ω ±5 percent for device types 06 and 07.
   R2 = 27Ω ±5 percent, R3 = 27Ω ±5 percent.
3. V_{CC} shall be high enough to insure that 5.0 V minimum is present at V_{CC} device terminal.

Device types 06 and 07



NOTES:

1. High temperature (175°C), condition F, burn-in circuit only.
2. R1 = 2 kΩ. R2 = 2Ω.
3. V_{CC} shall be chosen such that 5.0 V minimum is present at device terminal.

FIGURE 4. Burn-in and life test circuit - Continued.

Supersedes page 26 of MIL-M-38510/309D of 21 June 1985.

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